

Force

Force : Push or pull , Newton (N) $\approx$ the force it takes to hold up a small apple ($\sim 100g$) against the gravity

(kg·m/s²)

(kg)

(m/s²)

$$\text{Force} = \text{mass} \times \text{acceleration}$$

$$F = m a$$

Torque (2D)

Torque : a Twisting force , the rotational version of regular push/pull force

(N·m)

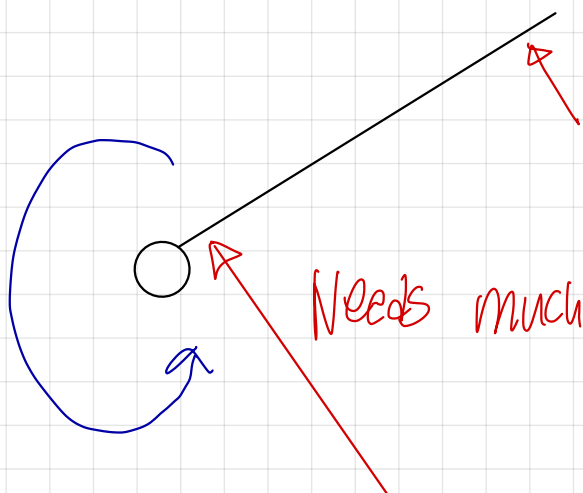
(N)

(m)

$$\text{torque} = \text{force} \times \text{distance from pivot}$$

= lever arm

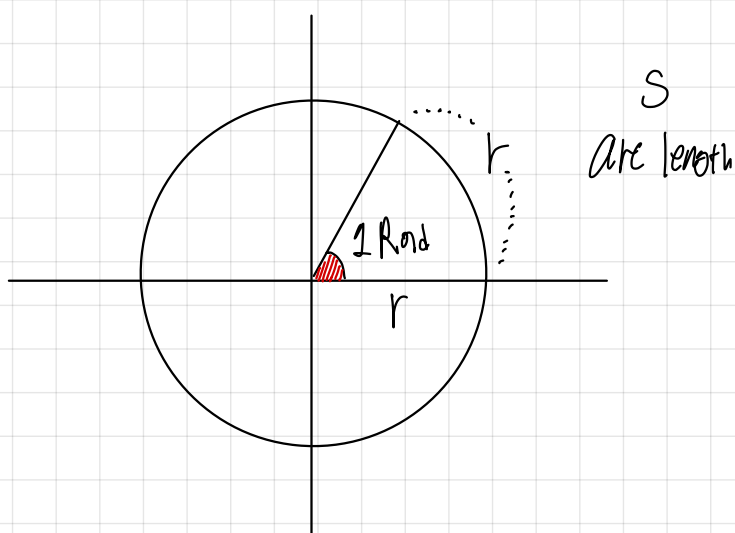
= Moment arm



$$I = F r$$

When the push is perpendicular to the lever

Radian



(m)

(m)

(unitless)

$$\text{Arc length} = \text{radius} \times \text{angle in radians}$$

$$s = r \theta$$

$$\theta = \frac{s}{r}$$

↳ A full circle : $360^\circ = 2\pi$ radians

$$\theta(\text{rad}) = \theta(\text{deg}) \times \frac{2\pi}{360}$$

$$\theta(\text{deg}) = \theta(\text{rad}) \times \frac{360}{2\pi}$$

↳ π = $\frac{\text{Circumference (m)}}{\text{Diameter (m)}}$ (unitless)

of ANY Circle

$$\pi = C/d \approx 3.14$$

↳ $C = \pi \times d = \pi \times 2r = 2\pi r$

$= (2\pi) r$

If arc length is C (the whole circle) then

$$S = \theta r \Rightarrow \theta = (2\pi)$$

Angular Velocity and Acceleration

$$\text{Velocity} = \frac{\Delta \text{position (m)}}{\Delta \text{time (s)}} \quad (\text{m/s})$$

$$V = \frac{\Delta x}{\Delta t}$$

How much distance per unit time

$$\text{Angular Velocity} = \frac{\Delta \text{angle (in rad)}}{\Delta \text{time (s)}} \quad (\text{rad/s})$$

$$\omega = \frac{\Delta \theta}{\Delta t}$$

How much angle per unit time

$$\text{Acceleration} = \frac{\Delta \text{velocity (m/s)}}{\Delta \text{time (s)}} \quad (\text{m/s}^2)$$

$$a = \frac{\Delta v}{\Delta t}$$

How fast velocity changes
per unit time

$$\text{Angular acceleration} = \frac{\Delta \text{angular velocity (rad/s)}}{\Delta \text{time (s)}} \quad (\text{rad/s}^2)$$

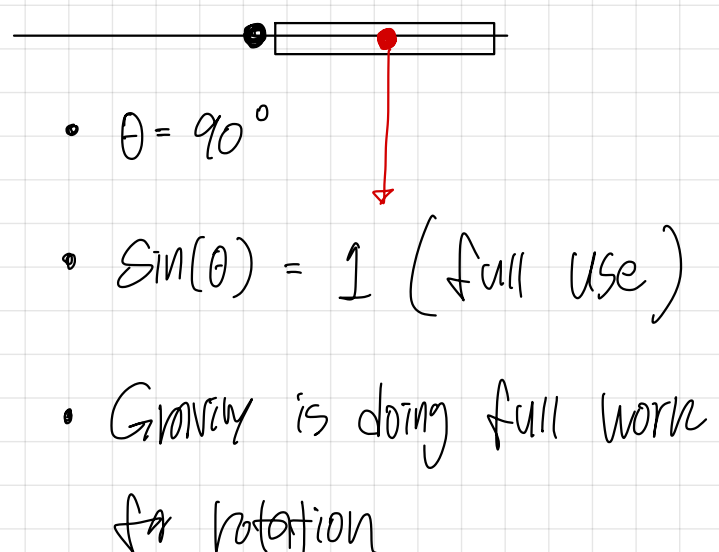
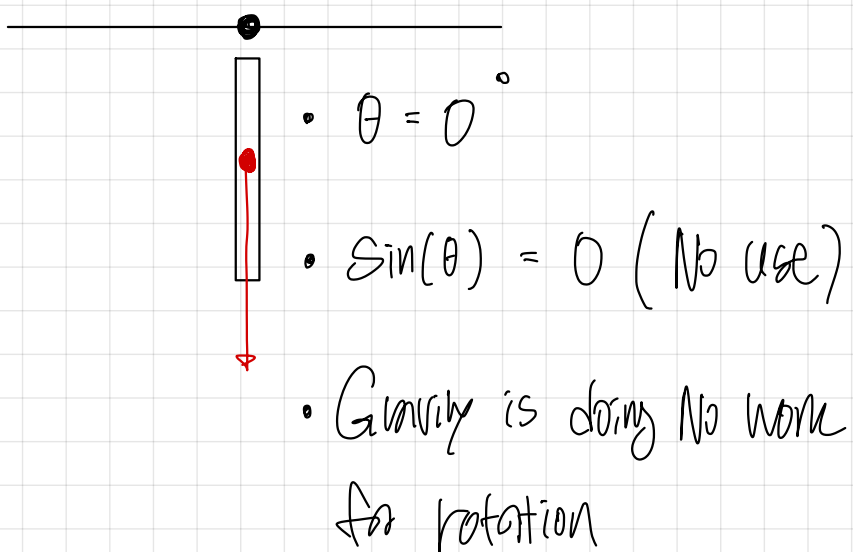
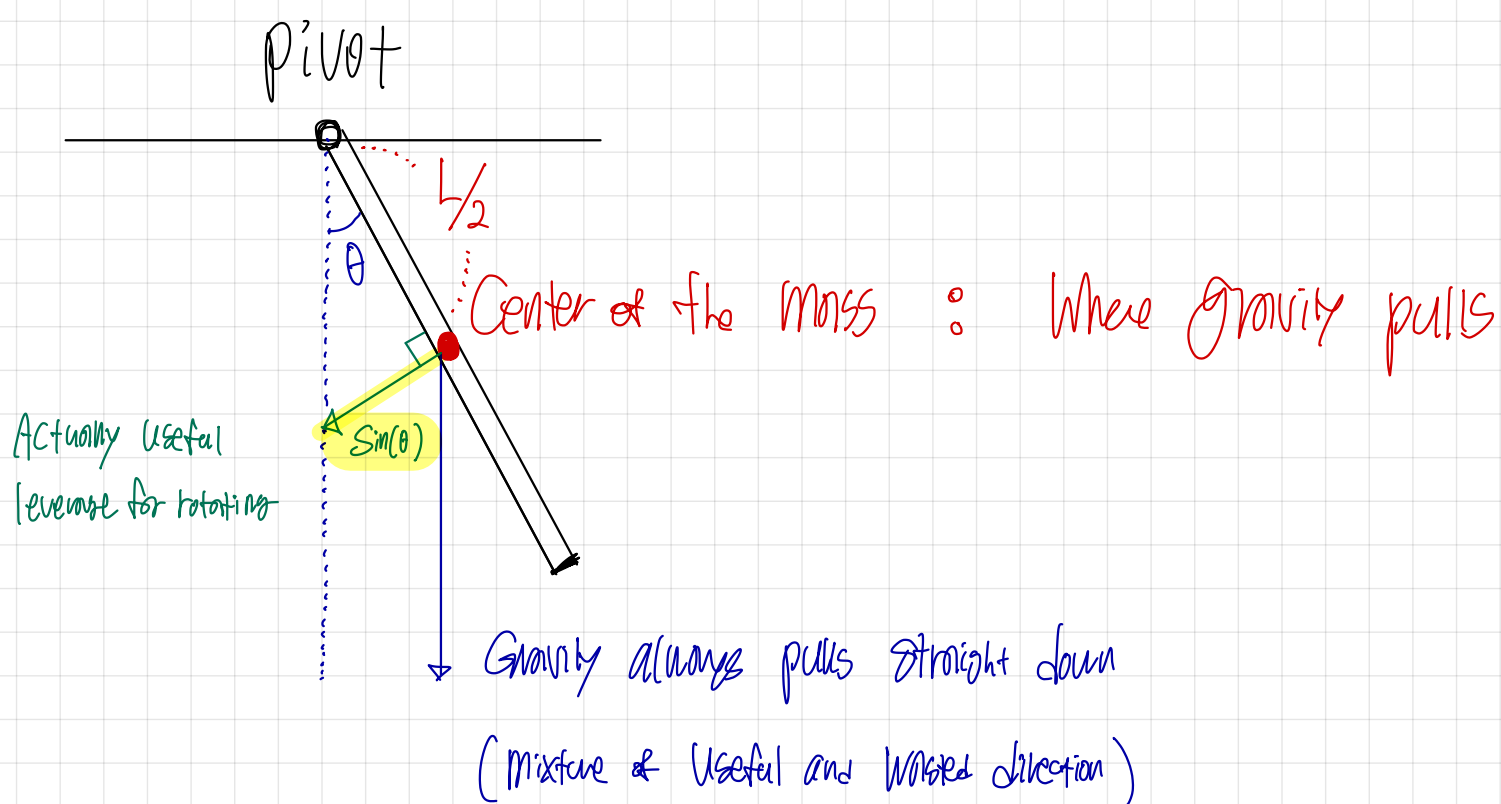
$$\alpha = \frac{\Delta \omega}{\Delta t}$$

How fast angular velocity changes
per unit time

Gravitational Torque

Gravitational Torque : the torque that gravity creates around the pivot trying to swing the pendulum

↳ torque $\tau = F \times r$ (moment arm)



Moment Arm : the perpendicular distance from the pivot to the line along with the force acts

$$\text{Moment arm} = \frac{L}{2} \times \sin(\theta)$$

(m)
(m)
(dimensionless)

$$I_{\text{gravity}} = F_{\text{gravity}} \times \text{Moment Arm}$$

(N·m)
(N)
(m)

$$= (m \times g) \times \left(\frac{L}{2} \times \sin(\theta) \right)$$

(kg)
(m/s²)
(m)
(unitless)

Damping Torque

Damping : Any force or torque that opposes the direction of current motion, grows in strength as the speed of motion increases, and is zero when there is no motion at all

1) Viscous damping (linear)

$$(N·m) \quad (N·m·s/rad) \quad (rad/s)$$

$$I = -C \times \omega$$

Smooth, proportional to speed, realistic for air resistance or fluid-line resistance in a joint

2) Coulomb Damping (dry friction or sticktion)

(N·m) (N·m) (unitless)

$$\tau = -\mu \times \text{Sign}(\omega)$$

A constant resistance, same strength no matter how fast

3) Quadratic Damping (air resistance at high speed)

(N·m) (N·m·s²/rad²) (rad/s) (rad/s)

$$\tau = -C \times \omega \times |\omega|$$

↳ τ_{total} for the pendulum model)

$$\tau_{\text{total}} = \tau_{\text{gravity}} + \tau_{\text{viscous}} + \tau_{\text{Coulomb}}$$

Inertia

Inertia : an Object's natural resistance to having its Motion Changed (Newton's first law)

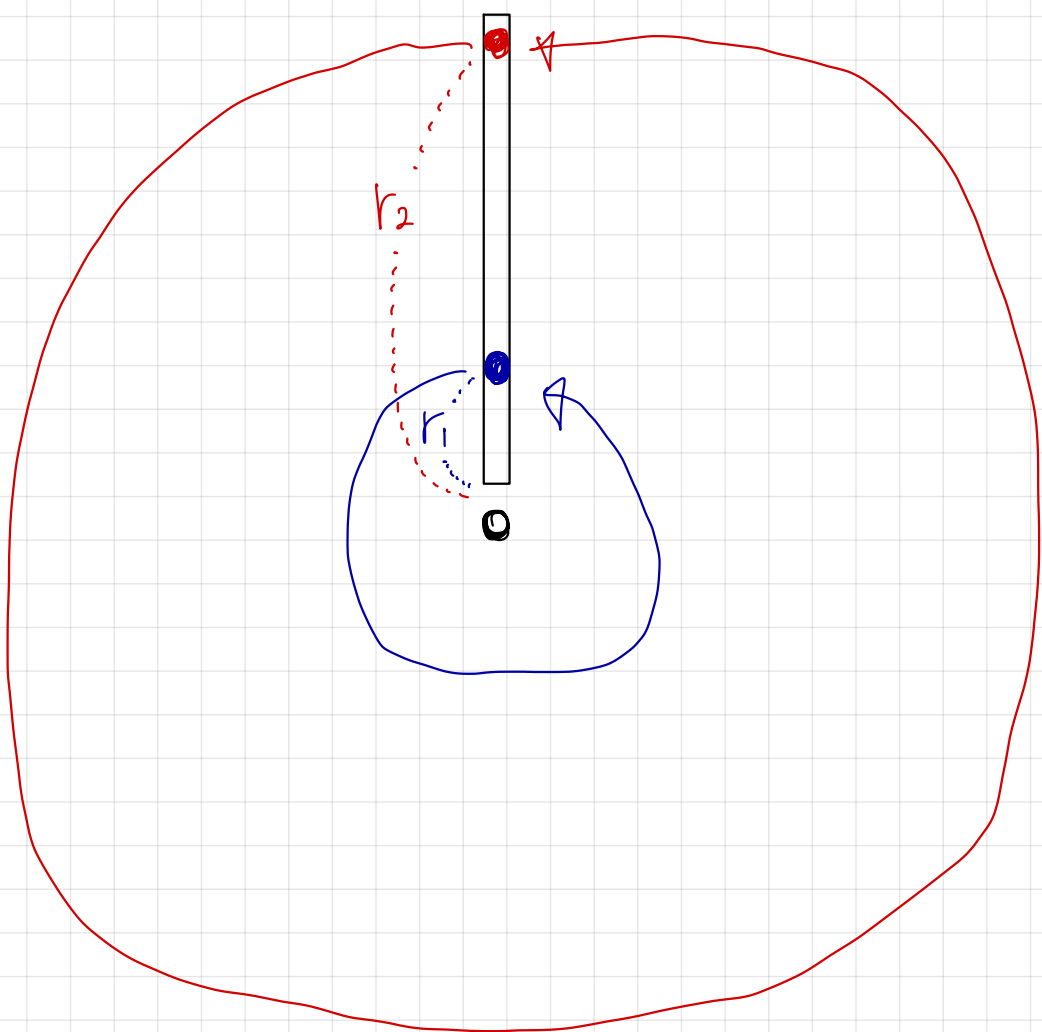
↳ Straight line motion : Inertia = Mass (kg)

$$F = \underline{m} a$$

Linear Motion (Translational motion) Not for Rotational Motion

↳ Why Chunks ?

Why Rotational Motion Needs something More than Mass?



Both points has the same amount of time to rotate

A point sitting far away from the pivot should travel much further thus require higher speed

⇒ more expensive to spin

$$(kg \cdot m^2) \quad (kg) \quad (m)$$

$$I = m \cdot r^2$$

0) If we push a mass on the end of a rod to make it spin faster, how much push is needed in terms of angular quantities?

$$1) \quad s = r \times \theta \quad \Rightarrow \quad \frac{ds}{dt} = r \times \underbrace{\frac{d\theta}{dt}}_{\omega : \text{angular velocity}}$$

↳ the distance if the mass sweeps through some angle θ

2) Velocity is defined as how fast distance changes over time

$$v = \frac{ds}{dt} = r \omega, \quad \frac{dv}{dt} = r \times \underbrace{\frac{d\omega}{dt}}_{\alpha : \text{angular acceleration}}$$

3) Acceleration : how fast velocity changes over time

$$a = \frac{dv}{dt} = r \alpha$$

4) Newton's second Law

$$F = m \times a = m \times r \times \alpha$$

5) Torque

$$L = F \times r = m \times r \times \alpha \times r = \underbrace{m r^2}_{I} \alpha$$

6) General Rule for Rotation (NZL for rotation)

Inertia angular acceleration

$$\therefore \tau_{\text{net}} = I \times \alpha$$

(N·m) (kg·m²) (rad/s²)

$$\hookrightarrow \tau_1 = m_1 \times r_1^2 \times \alpha$$

$$\tau_2 = m_2 \times r_2^2 \times \alpha$$

⋮

$$\tau_n = m_n \times r_n^2 \times \alpha$$

$$\tau_{\text{total}} = \tau_1 + \tau_2 + \dots + \tau_n = \alpha \times \underbrace{\sum m_i r_i^2}_I$$

(kg·m²)

(kg·m²)

$$\therefore I = \frac{\tau}{\alpha} = m r^2$$

↳ Discrete Chunks) $I = \sum m_j r_j^2$

Continuous Chunks) $I = \int_0^L r^2 dm$

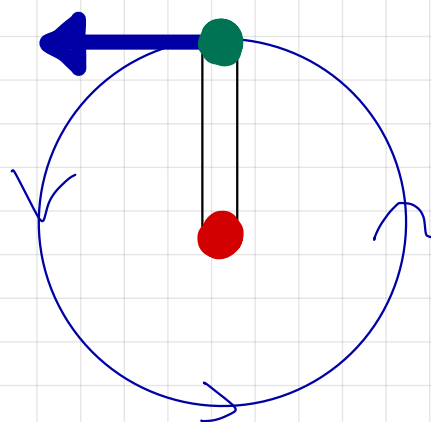
• Interpretation) $I = \tau / \alpha$

I is the amount of torque required to produce one unit of angular acceleration

⇒ If $I = 0.083 \text{ kg} \cdot \text{m}^2$ for the pendulum, to make this pendulum accelerate angularly at 1 rad/s^2 , we need exactly $0.083 \text{ N} \cdot \text{m}$ of torque

Inertia about the Pivot (end) across the rod

$$I = \frac{M \cdot L^2}{3} = \text{Rod mass} \cdot \text{Rod Length}^2 / 3$$



Derivation

i) mass per unit length

Total mass of the rod : M [kg]

Total length of the rod : L [m]

mass per unit length : m/L [kg/m]

ii) tiny slice

dr : a very small length

$\Rightarrow$ How much mass in that slice?

$dm = \text{mass per unit length} \times dr$

$dm = (m/L) \times dr \Rightarrow \text{one chunk mass}$

iii) Contribution

Contribution depends on the moment arm r

$$\text{Contribution} = dm \times r^2$$

$$= \left(\frac{m}{L} \right) dr \times r^2$$

↳ one slice's contribution

iv) Total Contribution (Since the origin to the end : 0 to L)

$$I = \int_0^L r^2 \left(\frac{m}{L} dr \right)$$

$$= \frac{m}{L} \int_0^L r^2 dr$$

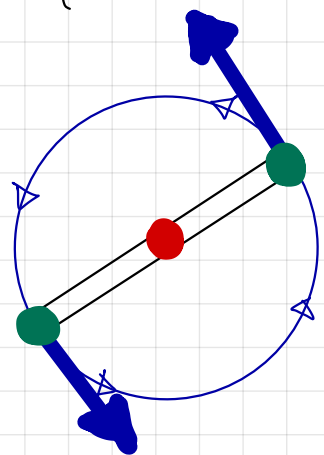
$$= \frac{m}{L} \left(\frac{r^3}{3} \Big|_{r=L} - \frac{r^3}{3} \Big|_{r=0} \right)$$

$$= \frac{m}{L} \cdot \frac{L^3}{3}$$

$$I = \frac{1}{3} m L^2$$

Inertia about the Center of Mass across the rod

How hard is it to change the rotational speed of the rod (when it spins around an axis) through the center of mass, perpendicular



$$\begin{array}{c} (\text{kg} \cdot \text{m}^2) \quad (\text{kg}) \quad (\text{m}) \\ I_{\text{com}} = \frac{m \cdot L^2}{12} = \text{rod mass} \cdot \text{rod length}^2 / 12 \end{array}$$

- How hard to change the rotational speed (def of Inertia) :

↳ means to speed up or down the spinning = Angular acceleration ($\alpha = \text{rad/s}^2$)

↳ how hard = how much torque ($\tau = I \cdot \alpha = (\text{N} \cdot \text{s}) = (\text{kg} \cdot \text{m/s}) \cdot (\text{rad/s}^2)$)

so Inertia is torque needed per unit of angular acceleration ($I = \frac{\tau}{\alpha}$)

ex) If $I_{\text{com}} = 0.0833 \text{ kg} \cdot \text{m}^2$ then to speed the rod's spin up by 1 rad/s^2 each second

we need $0.0833 \text{ N} \cdot \text{s}$ of torque

- When it spins about an axis :

↳ Axis of Rotation : the straight line that stays still while everything else turns around it

↳ With the torque fixed, a smaller moment of inertia gives a bigger angular acceleration

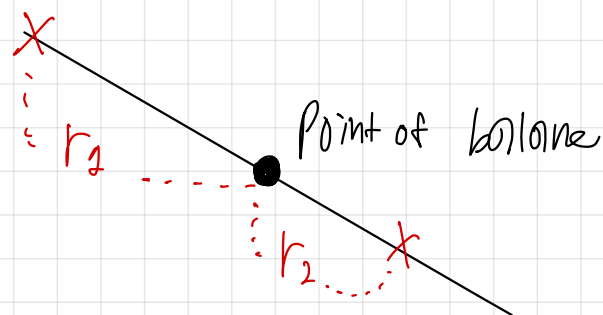
↳ arc length : $s = r \times \theta$

$(m) \quad (m)$
radius angle

- **Derivation**) a uniform rod, mass m (kg), length L (m),
axis through the center of mass perpendicular to the rod

1) inertia = $\int r^2 dm$ (r : the distance of each piece from the axis)

2) mass per unit length = m/L [kg/m]



mass of a slice : $dm = (m/L) dr$ [kg]

limits : r from $-L/2$ to $L/2$ [m] $\Rightarrow \int_{-L/2}^{L/2}$

↳ about the COM : r is measured from the balance point
about the end : r is measured from the end

3) Inertia) $I_{\text{com}} = \int_{-L/2}^{L/2} r^2 \cdot \left[(m/L) dr \right]$

$$= \frac{m}{L} \int_{-L/2}^{L/2} r^2 dr$$

$$= \frac{m}{L} \left[\frac{r^3}{3} \right]_{-L/2}^{L/2}$$

$$= \frac{m}{L} \left(\frac{(L/2)^3}{3} - \frac{(-L/2)^3}{3} \right)$$

$$= \frac{mL^3}{12}$$

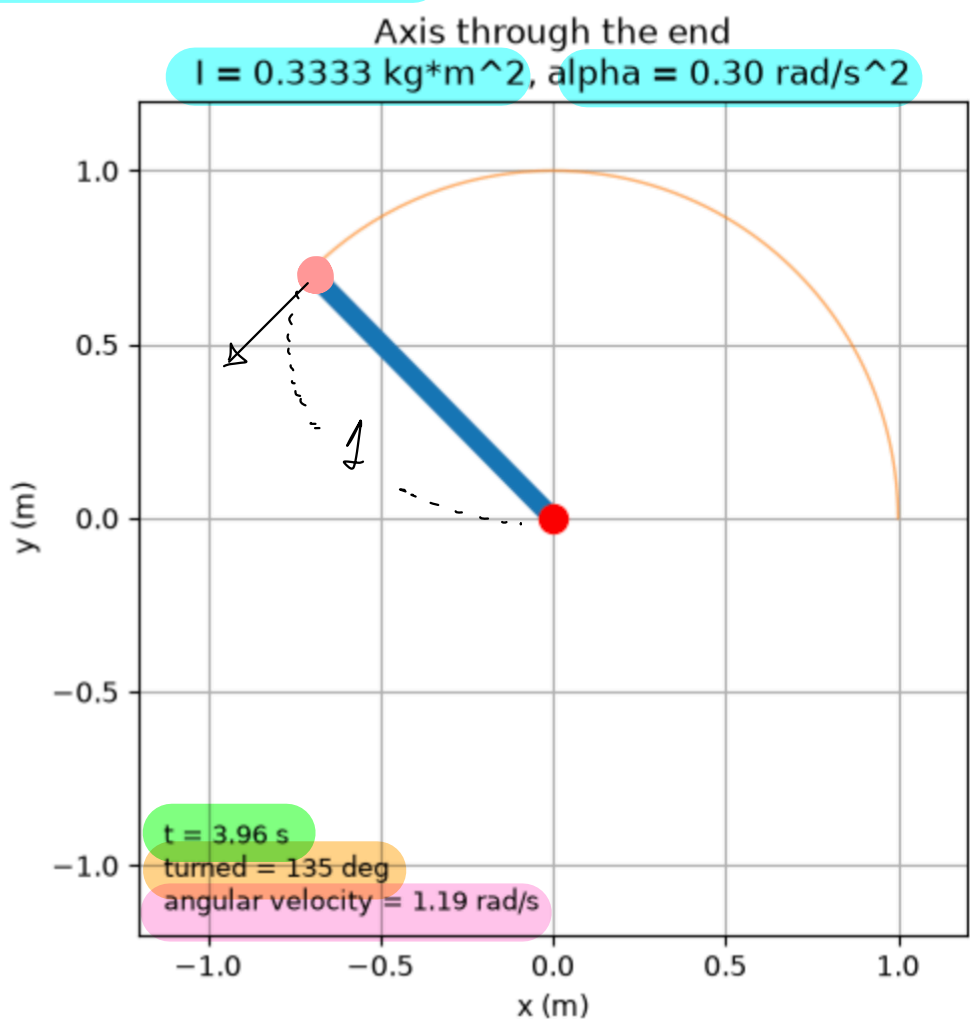
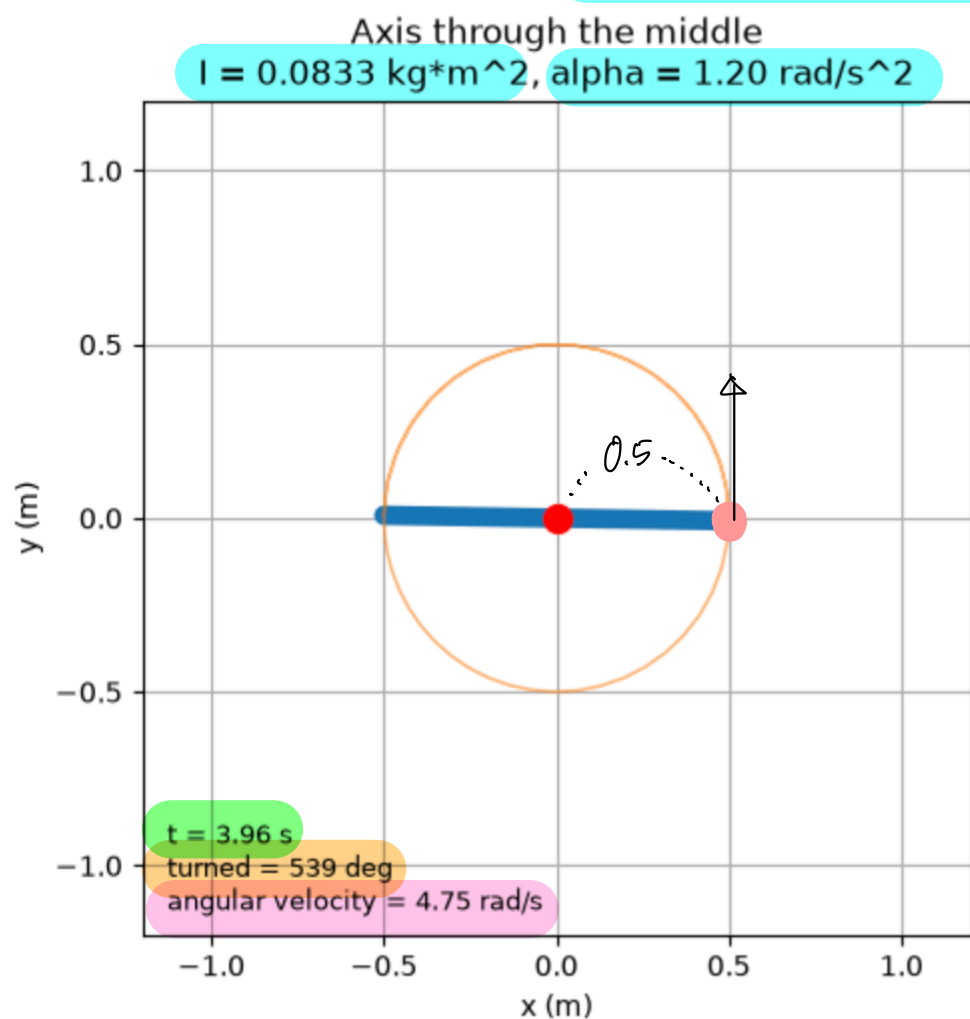
Linear speed/velocity of one point of the rod (tangential Speed Velocity):

$$\begin{matrix} \text{Speed} & \left(\frac{m}{s} \right) & (m) & \left(\frac{rad}{s} = \frac{1}{s} \right) \\ & V & = r \times \omega & = r \times \frac{\theta}{s} \\ & & \text{radius} & \text{Angular Velocity} \end{matrix}$$

⇒ Speed = distance traveled per unit time

⇒ Angular velocity $\omega = \text{Change in theta} / \text{Change in time}$

Same rod, same torque (0.1 N*m), different axis



$$\begin{matrix} r & \omega \\ \downarrow & \\ \text{Speed at the tip} & = (0.5) \times (4.75) \\ & = 2.4 \text{ m/s} \end{matrix}$$

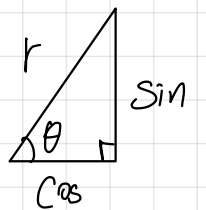
$$\begin{matrix} r & \omega \\ \downarrow & \\ \text{Speed at the tip} & = (1) \times (1.19) \\ & = 1.2 \text{ m/s} \end{matrix}$$

↳ the direction of V : Tangent to its circle (points along the circle right angle to the rod)

Why?)

The position of the tip when the rod has turned by angle θ (rad) with θ changing

at rate ω : $\text{Position} = (r \cdot \cos(\theta), r \cdot \sin(\theta))$



The Velocity which is the rate of change of position. Differentiate w.r.t time.

Since $\frac{d\theta}{dt} = \omega$: $\text{Velocity} = (-r \cdot \omega \cdot \sin(\theta), r \cdot \omega \cdot \cos(\theta))$

↳ $(r \cdot \sin(\theta))'$

↳ $(r \cdot \cos(\theta))'$

Check the Size : $\text{Speed} = \sqrt{(-r\omega \sin(\theta))^2 + (r\omega \cos(\theta))^2}$

$$= r \cdot \omega \cdot \sqrt{\sin^2 \theta + \cos^2 \theta}$$

$$= r \cdot \omega \cdot 1 = r \cdot \omega$$

Check the direction : Dot Product of Position Arrow and Velocity Arrow

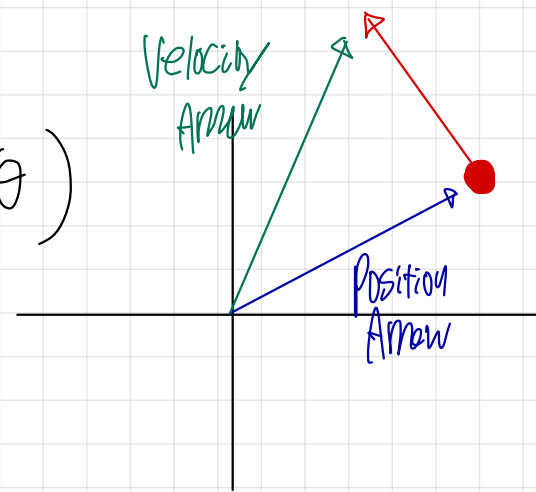
↳ dot product of $a = (a_x, a_y)$ and $b = (b_x, b_y)$

$$\underset{\text{dot}}{a \cdot b} = a_x b_x + a_y b_y \quad \text{Shows how aligned } a \text{ and } b$$

$$= (\text{length of } a) \times (\text{length of } b) \times \cos(\theta)$$

⇒ Position Arrow : $(r \cos \theta, r \sin \theta)$

⇒ Velocity Arrow : $(-r\omega \sin \theta, r\omega \cos \theta)$



$$\frac{d}{dt} \mathbf{PA} \cdot \mathbf{VA} = (r \cos \theta \cdot -r\omega \sin \theta) + (r \sin \theta \cdot r\omega \cos \theta)$$

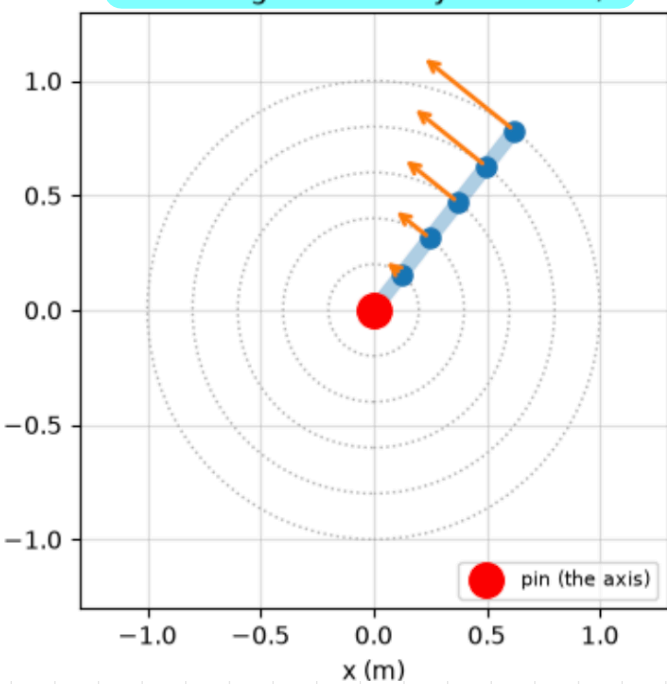
$$= -r^2 \omega \sin \theta \cos \theta + r^2 \omega \sin \theta \cos \theta$$

= 0 thus perpendicular to the position Arrow

Spinning about an axis: the pin does not move, every other point circles around it

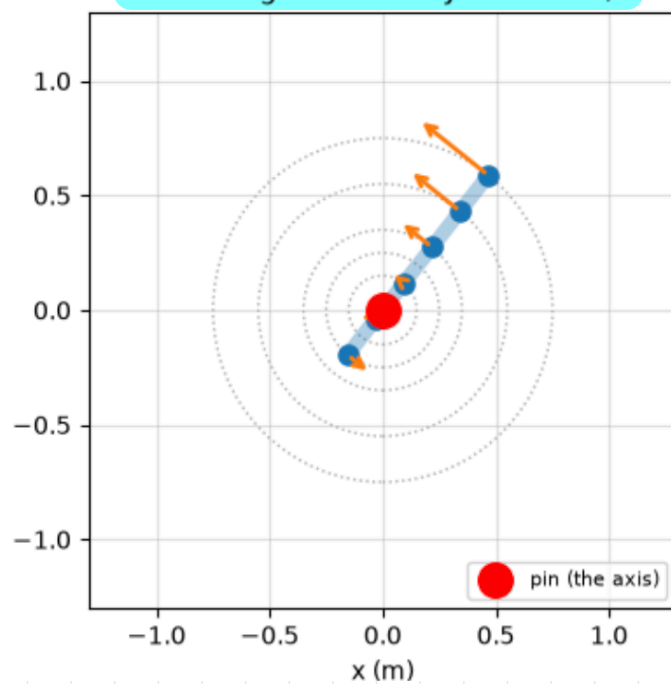
Pin at the end

same angular velocity = 1.5 rad/s



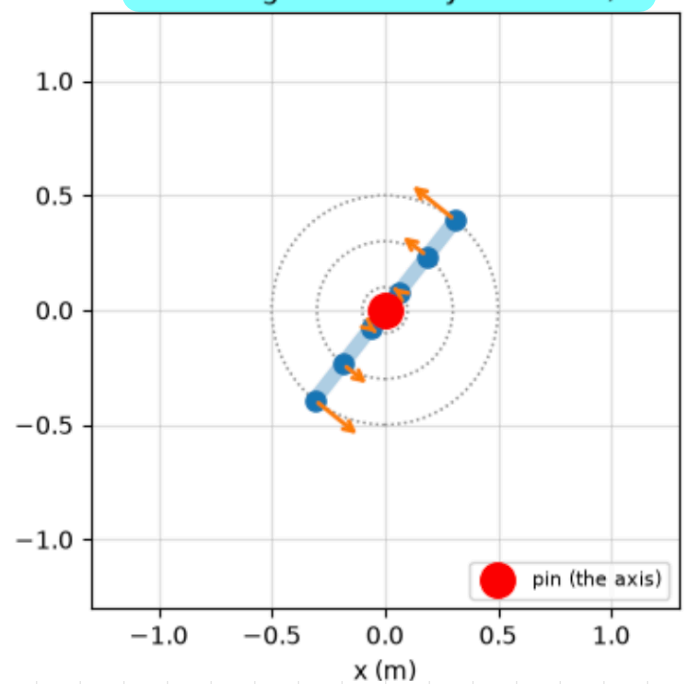
Pin a quarter of the way along

same angular velocity = 1.5 rad/s



Pin in the middle

same angular velocity = 1.5 rad/s

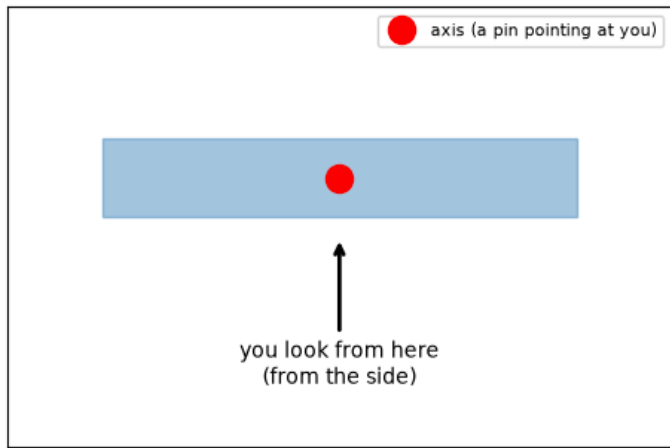


Rod Viewpoints

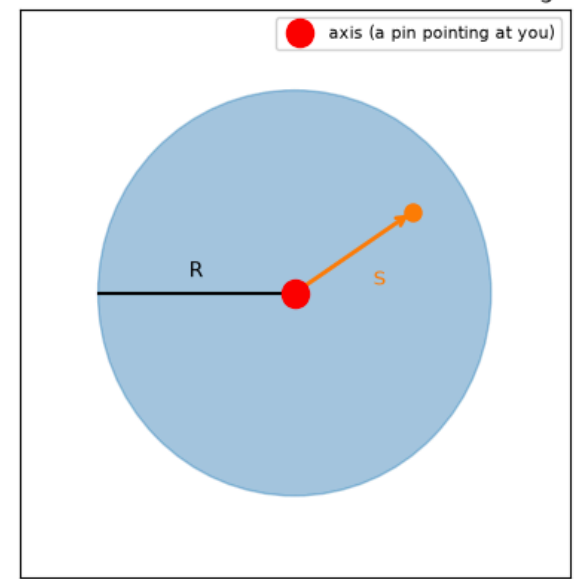
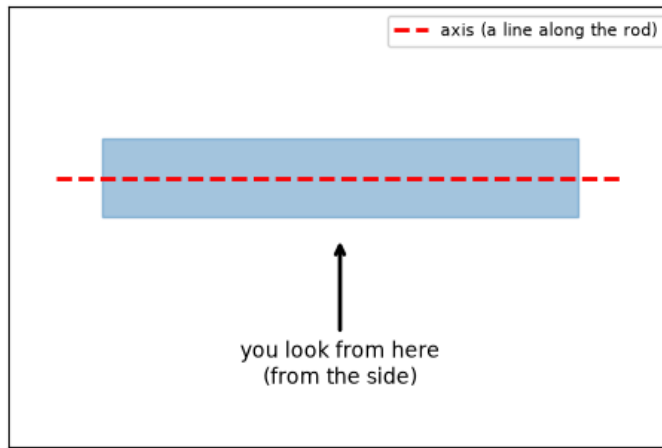
Same rod, different place to stand (not to scale: the rod is drawn thicker than it is)

(c) End view, looking along the axis used for the moment of inertia about the long axis

(a) Side view used for the moment of inertia about the center of mass



(b) The same side view, but the axis runs along the rod from here the axis is a LINE, not a dot



you look from here: the end of the rod, looking down its length

Side View

Side View

End View / Axial View
(Views along the axis)

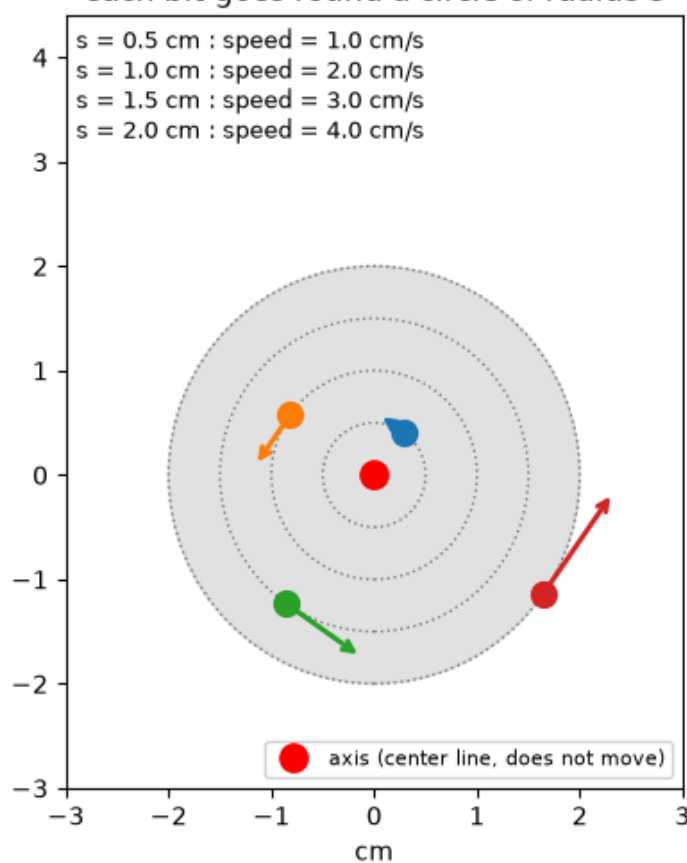
Inertia about the Z axis (I_{zz})

Principal Moment of Inertia (twirl)

↳ A number that says how stubborn the rod is against being
twirled about its own length

Twirling the rod about its own long axis, angular velocity = 2.0 rad/s

END VIEW (looking down the rod)
each bit goes round a circle of radius s



SIDE VIEW (same bits)
the rod stays put, each bit only moves up and down



(kg·m²) (kg) (m)

$$I_{zz} = \frac{m \cdot R^2}{2} = \text{Rod Mass} \cdot \text{Rod Radius}^2 / 2$$

• Derivation

1) Inertia : $I = \sum_i \Delta m_i r_i^2 = \int r^2 dm$

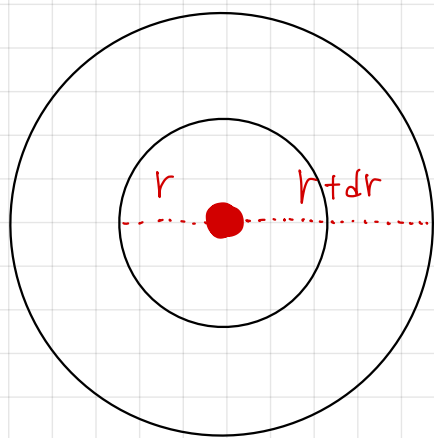
2) Mass per Unit Volume : R : rod radius , L : rod length , m : rod mass

Volume of the Cylinder $V = \pi R^2 L$ [m³]

Density $\rho = \frac{m}{V} = \frac{m}{\pi R^2 L}$ [kg/m³]

Since rod is uniform, the density is the same everywhere

3) Volume of one ring : dV



the rings lies between distance r and $r+dr$

and it is as long as the rod

so the volume is (area seen from the end) $\times L$

$$0 + \bigcirc + \bigcirc + \bigcirc + \dots = \text{concentric circles} = \text{shaded circle}$$

↳ we can control the size of R : start from $r=0$ to $r=R$

↳ Volume of One ring:

$$i) \pi(r+dr)^2 - \pi r^2 = 2\pi r dr + \pi dr^2$$

ii) dr is very small so dr^2 is negligible

$$dV = 2\pi r dr \cdot L \quad (m^3)$$

4) Mass of one ring: dm

$$dm = \rho dV \quad \left(\text{since } \rho = \frac{m}{V}\right)$$

$$= \rho \cdot 2\pi r dr \cdot L$$

$$= \frac{m}{\pi R^2 L} \cdot 2\pi r dr \cdot L$$

$$dm = \frac{2m}{R^2} r dr \quad [kg]$$

↳ All rings together should give m : $\int_0^R \frac{2m}{R^2} r dr = \frac{2m}{R^2} \frac{R^2}{2} = m$

5) Substitute

$$I_{zz} = \int_0^R r^2 dm$$

$$= \int_0^R r^2 \cdot \frac{2m}{R^2} r dr$$

$$= \frac{2m}{R^2} \int_0^R r^3 dr$$

$$= \frac{2m}{R^2} \left[\frac{r^4}{4} \right]_0^R$$

$$= \frac{2m}{R^2} \cdot \frac{R^4}{4}$$

$$I_{zz} = \frac{mR^2}{2} \quad [\text{kg} \cdot \text{m}^2]$$

Cheat Sheet

About the Center of Mass, across the rod : $dm = \frac{M}{L} dr$, $\int_{-L/2}^{L/2} \Rightarrow I = \frac{ML^2}{12}$

About the pivot (end), across the rod : $dm = \frac{M}{L} dr$, $\int_0^L \Rightarrow I = \frac{ML^2}{3}$

About the long axis

$$\therefore dm = \frac{2m}{R^2} r dr, \int_0^R \Rightarrow I = \frac{mR^2}{2}$$

Torque

Torque [N·m] is how strongly a push tries to twist an object about a pivot

Cross Product

$$(a_x, a_y, a_z) \quad (b_x, b_y, b_z)$$

$a \times b$ of two vectors a and b is a third vector that

- i) is perpendicular to both a and b
- ii) has a length $|a||b|\sin\theta$ where θ is an angle between a and b
- iii) points in the direction given by the right-hand rule
When the thumb points

↳ Order matters: a is the first and b the second

$$a \times b = -b \times a$$

Right Hand Rule

Inertia Tensor

A body can turn about three axes, and a torque about one axis can produce angular acceleration about another. We use 3×3 table for x, y, z axes.

$I_{\alpha\beta}$ = the torque about axis α that we need, per unit of angular acceleration about axis β

Inertia Tensor (I) : belongs to the rigid body and it converts the body's angular velocity ω [$1/s$] into its angular momentum L [$kg \cdot m^2/s$]

$$\underset{(3 \times 1)}{L} = \underset{(3 \times 3)}{I} \underset{(3 \times 1)}{\omega}$$

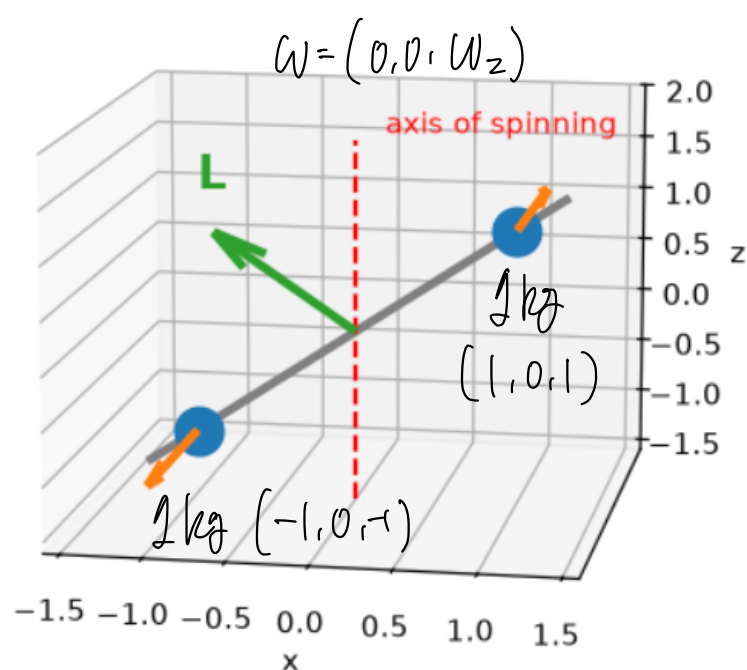
↳ 3D version of Momentum = mass $\times$ Velocity

Column = axis you spin about (ω : angular velocity)

row =
Which components
of L (or torque)
you look at

	x	y	z
x	I_{xx}	I_{xy}	I_{xz}
y	I_{yx}	I_{yy}	I_{yz}
z	I_{zx}	I_{zy}	I_{zz}

(a) Spin about z, but L is NOT along z
two 1 kg masses at $\pm(1, 0, 1)$ m



green L = angular momentum, orange = velocity of each mass (along y, in and out of the page)

↳ angular velocity $\omega = (\omega_x, \omega_y, \omega_z) \quad [\text{1/s}]$

position of one mass $r = (x, y, z) \quad [\text{m}]$

↳ velocity of one mass) $v = \omega \times r \quad [\text{m/s}]$

Setup) two masses of 1 kg

at $+(1, 0, 1)$ and $-(1, 0, 1)$

It spins about the z axis

With $\omega = (0, 0, \omega_z)$